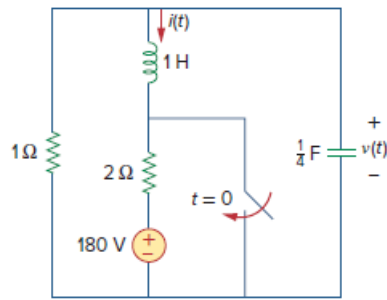


Problem

Given the circuit in Fig. 16.65, find $i(t)$ and $v(t)$ for $t > 0$.

Figure 16.65:



Step-by-step solution

Step 1 of 12

(Refer Figure 16.65 from textbook) Consider the switch is open for time $t < 0$. Then, replace the inductor by short circuit and the capacitor by open circuit.

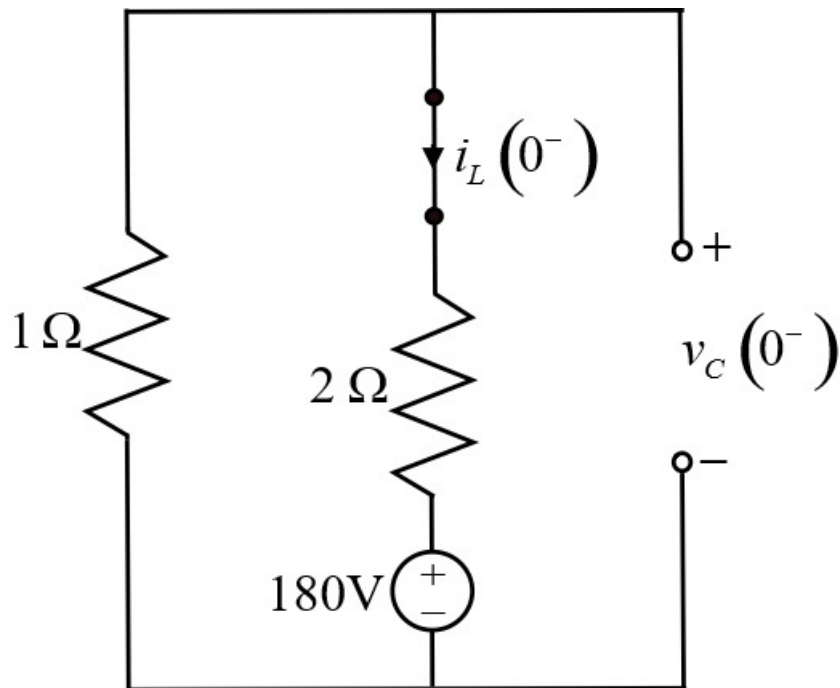


Figure 1

Step 2 of 12

Calculate the current $i_L(0^-)$ across the inductor.

$$\begin{aligned} i_L(0^-) &= \frac{-180}{1+2} \\ &= -60 \text{ A} \end{aligned}$$

Calculate the voltage across the capacitor.

$$\begin{aligned} v_C(0^-) &= \frac{180 \times 1}{1+2} \\ &= 60 \text{ V} \end{aligned}$$

Step 3 of 12

Consider the switch is close for time $t > 0$. Then, represent the inductor in s-domain.

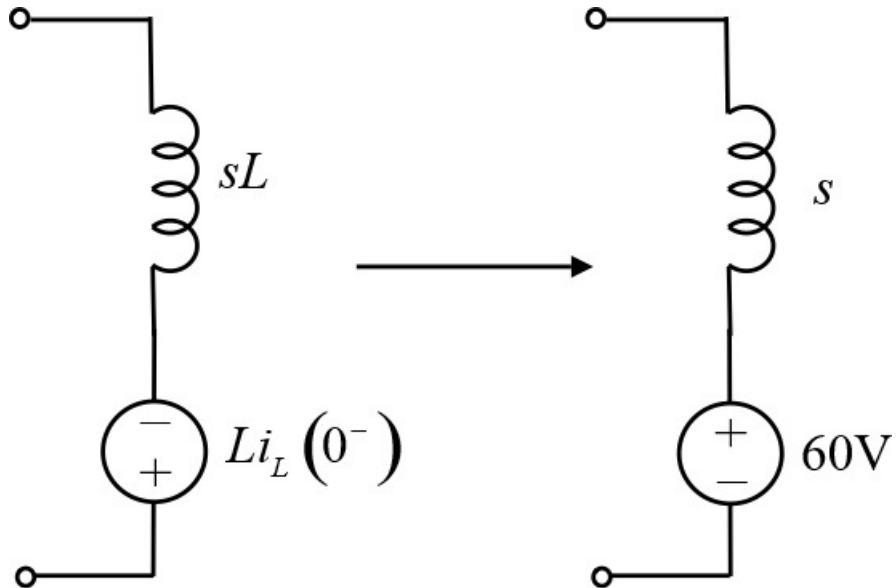


Figure 2

Step 4 of 12

Then, the complete circuit in s-domain is,

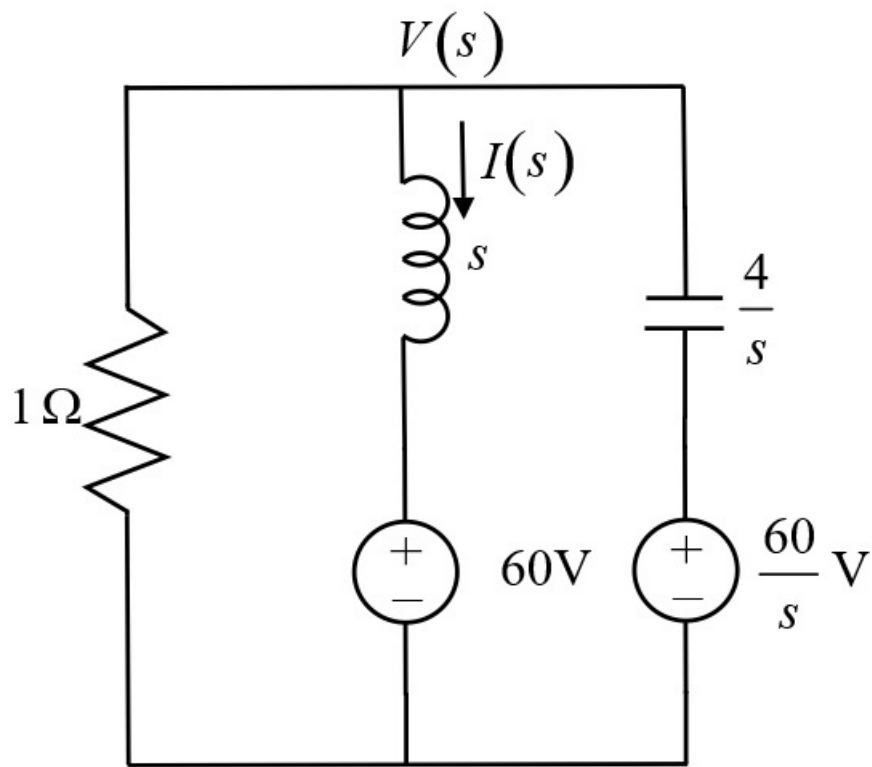


Figure 3

Step 5 of 12

Apply the nodal analysis in Figure 3.

$$\frac{V(s)}{1} + \frac{V(s) - 60}{s} + \frac{V(s) - \left(\frac{60}{s}\right)}{\left(\frac{4}{s}\right)} = 0$$

$$\frac{V(s)}{1} + \frac{V(s) - 60}{s} + \frac{sV(s) - 60}{4} = 0$$

$$\frac{4sV(s) + 4V(s) - 240 + s^2V(s) - 60s}{4s} = 0$$

$$V(s)(s^2 + 4s + 4) = 60s + 240$$

Rewrite the above equation.

$$V(s) = \frac{60s + 240}{(s^2 + 4s + 4)}$$

$$= \frac{60s + 240}{(s + 2)^2} \dots\dots (1)$$

Step 6 of 12

Do the partial fraction of equation (1).

$$\frac{60s + 240}{(s + 2)^2} = \frac{A}{(s + 2)} + \frac{B}{(s + 2)^2} \dots\dots (2)$$

$$60s + 240 = A(s + 2) + B$$

$$60s + 240 = sA + 2A + B$$

Compare the coefficient of s^1 and s^0 .

$$A = 60$$

$$B = 120$$

Step 7 of 12

Substitute 60 for A and 120 for B in equation (2).

$$\frac{60s + 240}{(s + 2)^2} = \frac{60}{(s + 2)} + \frac{120}{(s + 2)^2}$$

Rewrite the above equation.

$$V(s) = \frac{60}{(s + 2)} + \frac{120}{(s + 2)^2} \dots\dots (3)$$

Step 8 of 12

The inverse Laplace transform of $\frac{1}{s^2}$ is,

$$\frac{1}{s^2} \leftrightarrow t$$

The shifting property of inverse Laplace transform is,

$$\frac{1}{s + a} \leftrightarrow e^{-at}$$

Take the inverse Laplace transform of equation (3).

$$v(t) = [60(1 + 2t)e^{-2t}]u(t)$$

Hence, the voltage across the capacitor is $\boxed{[60(1 + 2t)e^{-2t}]u(t) \text{ V}}$.

Step 9 of 12

Calculate the current $I(s)$ across the inductor.

$$\begin{aligned} I(s) &= \frac{V(s) - 60}{s} \\ &= \frac{\left(\frac{60s + 240}{(s + 2)^2} - 60 \right)}{s} \\ &= \frac{60s + 240 - 60(s^2 + 4s + 4)}{s(s + 2)^2} \\ &= \frac{-60s^2 - 180s}{s(s + 2)^2} \end{aligned}$$

Rewrite the above equation.

$$I(s) = \frac{-60s - 180}{(s + 2)^2} \dots\dots (4)$$

Step 10 of 12

Do the partial fraction of equation (4).

$$\frac{-60s - 180}{(s + 2)^2} = \frac{A}{(s + 2)} + \frac{B}{(s + 2)^2}$$

$$-60s - 180 = As + 2A + B$$

Compare the coefficient of s^1 and s^0 .

$$A = -60$$

$$B = -60$$

Step 11 of 12

Substitute -60 for A and B .

$$\frac{-60s - 180}{(s + 2)^2} = \frac{-60}{(s + 2)} - \frac{60}{(s + 2)^2}$$

Rewrite the above equation.

$$I(s) = \frac{-60}{(s + 2)} - \frac{60}{(s + 2)^2} \dots\dots (5)$$

Step 12 of 12

The inverse Laplace transform of $\frac{1}{s^2}$ is,

$$\frac{1}{s^2} \leftrightarrow t$$

The shifting property of inverse Laplace transform is,

$$\frac{1}{s + a} \leftrightarrow e^{-at}$$

Take the inverse Laplace transform of equation (5).

$$i(t) = [-60(1 + t)e^{-2t}]u(t)$$

Hence, the current across the inductor is $\boxed{[-60(1 + t)e^{-2t}]u(t) \text{ A}}$.

